

Mihaela Cătălina Stoian

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Interests

Neuro-Symbolic AI • Knowledge-Aided ML • Deep Generative Models • Large Language Models

I build neuro-symbolic AI systems that inject background-knowledge requirements into neural networks during training and *guarantee* those requirements are satisfied. My focus is making these knowledge-aided frameworks practical for real-world use — synthesising tabular data, autonomous driving, and constraining large language models for theorem proving.

Education

University of Oxford

DPhil in Computer Science

Oxford, UK

2021–2026

- **Thesis:** Knowledge-Refined Deep Neural Networks for Real-World Applications
- **Supervisor:** Prof. Thomas Lukasiewicz
- Developed the **first frameworks** that make deep generative models for tabular data provably compliant with background knowledge — expressed as linear constraints and disjunctions of linear inequalities capturing non-convex and even disconnected feasible spaces.
- **Areas:** deep learning, generative modelling, autonomous driving, neuro-symbolic AI

The University of Edinburgh

MInf (Master of Informatics), First Class Honours

Edinburgh, UK

2014–2019

- **Master's thesis:** Improving extrinsic performance in zero-resource settings
- **Bachelor's thesis:** Multilingual bottleneck features for zero-resource speech processing
- **Supervisor:** Prof. Sharon Goldwater
- **Areas:** NLP, speech-to-text machine translation, speech processing.

Work Experience

Imperial College London

Research Associate

London, UK

2025–present

- **PI:** Dr. Eleonora Giunchiglia
- **Autoformalisation:** building methods to constrain language models so they translate informal mathematics into formal statements verifiable by proof assistants such as Lean.
- Funded by **Renaissance Philanthropy**.

FiveAI

Research Intern — Computer Vision & Autonomous Vehicles

Oxford, UK

2020–2021

- **Supervisor:** Dr. Tommaso Cavallari
- **3D symmetry detection:** designed a method that estimates planar reflective symmetries on 3D inputs by slicing along the height dimension and feeding slices to a 2D convolutional recurrent regressor, avoiding costly 3D convolutions.
- Published at the CVPR 3DVR Workshop 2021.
- **Patent:** *3D Perception*, US 20240212189 A1 (Five AI Ltd, 2024; pending).

The University of Edinburgh

Research Assistant — NLP & Machine Translation

Edinburgh, UK

2019

- **Supervisor:** Prof. Sharon Goldwater
- **Speech-to-text translation:** showed that ASR pretraining with target-language data augmentation improves translation performance, and that pretrained-ASR word error rate directly predicts translation quality.
- Published at ICASSP 2020.

ETH Zurich

Zurich, Switzerland

Summer Research Fellow — Network Verification & Program Synthesis

2018

- **Supervisors:** Prof. Martin Vechev, Asst. Prof. Dana Drachler Cohen
- **Program synthesis for P4:** built a Python parser for P4 (Programming Protocol-Independent Packet Processors) and used the Z3 solver to encode and synthesise intended packet-processing behaviour.

Technical Skills

Languages — Python (*proficient*); C++, Java, Bash/Shell (*familiar*)

Deep learning — PyTorch, PyTorch Lightning, TensorFlow; Hugging Face, verl

ML & data — scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Seaborn; Jupyter

Tooling — Linux/command line, Git, Docker, Slurm, Conda, Weights & Biases, TensorBoard, $\LaTeX$

Software & Patents

Software: PiShield — a PyTorch framework for integrating background-knowledge requirements into neural networks [github.com/mihaela-stoian/PiShield]

Patent: *3D Perception*, US 20240212189 A1 (Five AI Ltd, 2024; pending)

Publications

- [1] J. Ong Jun Leang, Y. Zhao, **M. C. Stoian**, W. Li, S. B. Cohen, E. Giunchiglia. Can I Have Your Order? Monte-Carlo Tree Search for Slot Filling Ordering in Diffusion Language Models. *ICML* 2026.
- [2] **M. C. Stoian**, T. Lukasiewicz, E. Giunchiglia. A Survey on Deep Learning Approaches for Tabular Data Generation: Utility, Alignment, Fidelity, Privacy, Diversity, and Beyond. *TMLR* 2026.
- [3] **M. C. Stoian**, E. Giunchiglia. Beyond the Convexity Assumption: Realistic Tabular Data Generation under Quantifier-Free Real Linear Constraints. *ICLR* 2025.
- [4] **M. C. Stoian**, S. Dyrnishi, M. Cordy, T. Lukasiewicz, E. Giunchiglia. How Realistic Is Your Synthetic Data? Constraining Deep Generative Models for Tabular Data. *ICLR* 2024.
- [5] **M. C. Stoian**. Deep Learning with Requirements in the Real World. *IJCAI* Doctoral Consortium 2024.
- [6] S. Dyrnishi, **M. C. Stoian**, E. Giunchiglia, M. Cordy. Deep Generative Models as an Adversarial Attack Strategy for Tabular Machine Learning. *ICMLC* 2024.
- [7] **M. C. Stoian**, A. Tatomir, T. Lukasiewicz, E. Giunchiglia. PiShield: A PyTorch Package for Learning with Requirements. *IJCAI* 2024.
- [8] E. Giunchiglia, A. Tatomir, **M. C. Stoian**, T. Lukasiewicz. CCN+: A Neuro-Symbolic Framework for Deep Learning with Requirements. *Int. J. Approx. Reason.* 171, 2024.
- [9] S. Khan, I. Teeti, R. J. Alitappeh, **M. C. Stoian**, E. Giunchiglia, G. Singh, A. Bradley, F. Cuzzolin. ROAD-Waymo: Action Awareness at Scale for Autonomous Driving. *arXiv:2411.01683*, 2024.
- [10] **M. C. Stoian**, E. Giunchiglia, T. Lukasiewicz. Exploiting T-norms for Deep Learning in Autonomous Driving. *NeSy* 2023.
- [11] E. Giunchiglia, **M. C. Stoian**, S. Khan, F. Cuzzolin, T. Lukasiewicz. ROAD-R: The Autonomous Driving Dataset with Logical Requirements. *Machine Learning* 112, 2023.
- [12] E. Giunchiglia, **M. C. Stoian**, T. Lukasiewicz. Deep Learning with Logical Constraints. *IJCAI* 2022.
- [13] **M. C. Stoian**, T. Cavallari. Recurrently Estimating Reflective Symmetry Planes from Partial Pointclouds. *CVPR Workshop on 3D Vision and Robotics* 2021.
- [14] **M. C. Stoian**, S. Bansal, S. Goldwater. Analyzing ASR Pretraining for Low-Resource Speech-to-Text Translation. *ICASSP* 2020.

Awards

Oxford PhD Runner-up Prize , G-Research	2025
EPSRC Scholarship for Doctoral Studies , University of Oxford	2021–2025
Women in Quant Finance Grant , G-Research	2025
IJCAI Grant , IJCAI-AIJ	2024
Conference Travel Grants , St Hilda's College, Oxford	2023, 2024
Best Paper Award , AI4AD Workshop @ IJCAI	2022

Selected Talks

OxBridge Women in CS Conference , University of Oxford (<i>selected for oral presentation</i>)	2026
Civil Service Leadership Group , Imperial College London (<i>invitation-only</i>) <i>One of only four projects selected; presented to senior UK civil servants, incl. the Head of the Civil Service.</i>	2025
Dagstuhl Seminar on Logic and Neural Networks (<i>invitation-only</i>)	2025
IJCAI Doctoral Consortium	2024
SnT , University of Luxembourg (<i>invited</i>) <i>Security, Reasoning and Validation reading group.</i>	2024
Sony AI (<i>invited</i>) <i>Barcelona and Tokyo reading groups.</i>	2024

Mentoring

Tutorials

Beyond Soft Penalties: Hard Constraints for Neural Networks with Continuous Outputs — UAI 2026 (*forthcoming*)
Co-presented; introduces methods for enforcing hard constraints over continuous neural network outputs.

Co-supervision

A. Godun — MSc, TU Wien (current)
S. Lee — MSc, Imperial College London (2025)
L. Pejic — BSc, TU Wien (2025)

Teaching

— Class Tutor, University of Edinburgh (2017–2019)

Discrete Mathematics & Mathematical Reasoning
Algorithms, Data Structures & Learning
Processing Formal & Natural Languages

Service

Workshop & challenge organiser

ROAD++: The Third Workshop & Challenge — ECCV 2024
ROAD-R: The Road Event Detection with Requirements Challenge — NeurIPS 2023
ROAD++: The Second Workshop & Challenge — ICCV 2023

Reviewing

Conferences — NeurIPS, ICLR, ICML, IJCAI, NeSy
Journal — *Machine Learning*
Workshops — RepL4NLP @ ACL 2022, NeSy-GeMs @ ICLR 2023

Languages

English (*proficient*) • Romanian (*native*) • German (*elementary*)